



Radiative transfer, absorption, and reflection by metal powder beds in laser powder-bed processing

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ABSTRACT

Laser powder-bed processing is widely employed in 3D printing. The technological processes strongly depend on such effective radiative-transfer properties of powder beds as the absorptance and the radiation penetration depth. However, it is still not completely clear what are the principal factors influencing these properties and what are the possible ranges of their variation. For theoretical investigation of the radiative transfer, a ray-tracing method is developed where the effective properties of powder beds are expanded in series of the reflectance of the solid-phase surface. Once the coefficients of the power series are calculated, they can be used for any combination wavelength/material thus facilitating parametric analysis. Powder beds are modeled by equal spheres packed in regular cubic-symmetry structures of various density. The calculated radiation penetration depth can vary from few particle diameters for close-packed structures with moderate solid-phase absorptance to few tens of particle diameters for less dense power beds of a highly reflecting material. The effective absorptance of the powder bed is revealed to decrease significantly with the packing density. The calculated dependence of the powder-bed absorptance versus the solid-phase absorptance and the density is validated by the known experimental data. The calculated angular distribution of the radiation reflected by a powder bed slightly deviates from the cosine-like distribution corresponding to the diffuse reflection law. It is validated by the known experimental data. A modified model of equivalent medium is proposed to approximate the obtained ray tracing results by analytic expressions.

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1. Introduction

Radiative transfer in packed particle beds attracts the attention of researchers in various fields including nuclear [1], solar [2], and hydrogen [3] energetics and thermal insulation [4]. The goal of these experimental and theoretical studies is to estimate the radiative thermal conductivity and the reflectance/absorptance/transmittance properties important for the applications. Theoretical approaches of view-factor matrix [1], radiation distribution factors [5], neural networks [6], and generalized radiative transfer equation [7] have been developed. The Monte Carlo ray tracing (MCRT) is widely employed in these approaches. It requires generation of digital models of the particle beds, which is the important part of the whole radiative-transfer model. The most useful models of particle beds are random packing of spheres by the discrete element method [1,5] and regular periodic structures of spheres [1,3].

Laser processing of powder beds is finding ever-widening application in engineering due to a rapid development of 3D printing employing metal powder bed processes. Light absorption is the

primary interaction of an incident laser beam with a powder bead, which can induce other physical processes important for a specific technology. In 2000, Tolochko et al. [8] accomplished a comprehensive experimental measurement of normal-hemispherical reflectance for various metal powders at the wavelengths of 1.06 and 10.6 μm that are the most employed ones in technological lasers. They found that a powder bed is considerably more absorbing than a flat surface of the same material. Later, measurements of normal-hemispherical reflectance [9,10] and absorptance at normal incidence [11] in other scientific groups confirmed the results of Tolochko et al. and widened the list of studied materials and wavelengths.

High-quality 3D printing requires obtaining thin uniform powder layers by mechanical deposition. The best conditions for this are generally attained when powder particles are spherical and the gravity force applied to a particle is comparable with or greater than the Van der Waals adhesion forces between particles [12]. The latter condition defines an inferior bound for particle size depending on material. Commercial metal 3D printers commonly employ powders of spherical particles around 20 microns or greater in diameter and lasers with 1.07 μm wavelength [13]. The great diame-

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ter/wavelength ratio allows one to use the ray optics for explaining light absorption in the powder beds. The most rigorous theoretical results were obtained by MCRT.

Wang et al. [14,15] developed an MCRT model to calculate absorptance of an array of randomly distributed non-intersecting polydisperse spheres. For the specular reflection law, this model satisfactorily agrees with experimental data. Boley et al. [16] calculated the absorptance of three model arrays of spheres: a bed of two close-packed layers of equal spheres and two beds of polydisperse spheres with a Gaussian and a bimodal distributions. They revealed a dependence of the effective absorptance of the powder bed on its structure model accepted for calculations, which is especially important for highly reflective materials. In view of this fact, it is not surprising that the values of the effective absorptance reported by Boley et al. are considerably less than the corresponding values reported by Wang et al. Zhang et al. [10] compared the close-packed structure of equal spheres with denser structures obtained by filling the pores between the spheres with additional smaller spheres. Their MCRT calculations did not find a clear dependence of the effective absorptance on the density.

Another theoretical approach is the model of effective absorbing scattering medium that reduces the problem to the radiative transfer equation (RTE). The parameters of this model can be estimated from the parameters of the powder bed. Gusarov and Kruth [17] solved a half-space problem for RTE and found the effective absorptance of the powder bed as function of the absorptance of the flat surface. In this approximation, the effective absorptance is independent of powder bed density, which contradicts the recent MCRT results [16]. Surprisingly, the RTE model agrees well with experimental data, see [18].

Not only the effective absorptance but also the penetration profile and volumetric absorption affect laser processing of powder beds. Transmittance of powder beds was measured [19] to estimate volumetric light absorption. Both the MCRT and RTE models indicate that the powder bed structure significantly influence the penetration depth. Normal-directional reflectance measured in [9] can provide additional information about light interaction with powder beds. However, the analysis of these data using an RTE model [9] did not result in clear conclusions. It was probably due to restrictions of the model.

The weak point of the existing ray-tracing studies of radiative transfer in powder beds is modeling the realistic structure of the beds. There are known works aiming to simulate mechanical deposition of powders in the conditions typical for industrial laser powder-bed machines, for example [12], but the known theoretical investigations of radiation transfer do not use such realistic models.

Thus, Wang and Kruth [14] generated their packed-bed models by sequential random placing of particles until there is a free space for a particle. They did not communicate the resulted packing density. Boley et al. [16] and Kovalev and Gusarov [20] used the sequential rain model for random packing. Boley et al. did not communicate the packing density too. Kovalev and Gusarov considered the adhesion Van der Waals force in addition to the gravity force. Therefore, the packing density varied with the particle size because of varying the adhesion/gravity force ratio [20]. However, one cannot be sure if such variation is realistic because the rain model considerably differs from the blade spreading typically used in laser powder-bed processes. Boley et al. [16] and Zhang et al. [10] studied close-packed regular structures of spheres but their density was significantly greater than typical powder densities experimentally measured by Gusarov et al. in the range from 0.33 to 0.67 [9].

Kovalev and Gusarov [20] found a considerable anisotropy in their model packed beds. It is not clear if one can expect a significant anisotropy in realistic powder beds. Wang and Kruth [14] and

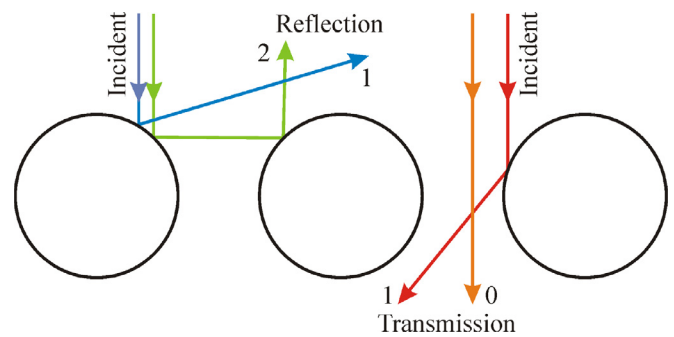


Fig. 1. Reflection and transmission of collimated incident radiation by an array of opaque specularly-reflecting spheres. The numbers near the exiting rays designate the number of reflections from spheres.

recently Kovalev et al. [21] reported the ray tracing results indicating that incident radiation penetrates into packed beds to the depth of about few particle sizes. Gusarov and Kruth [17] explained the penetration in the framework of the RTE model and found that the principal parameter responsible for the penetration depth is the specific surface. These results suggest that the arrangement of particles on the surface of the packed bed is also important for reflection and penetration. Zarrouati et al. [22] reported considerable periodic variations of packing density near walls of a container with a packed bed. Similar irregularities are possible near the top surface too.

The gravity force increases as the particle size cubed while the adhesion Van der Waals force between particles is proportional to the particle size [23]. That is why the adhesion force is negligible in packed beds of macroscopic particles and becomes comparable with or greater than the gravity force in powder beds. This fact essentially determines the lower density of powders [12,20]. Experimentally measured packing density of powders [9] varies depending on the material, particle size and shape. Generally, the density is less for finer powders and powders of irregular particles [9]. Prediction of powder bed properties in a wide range of packing density is a challenge. The measured reflectance of powder beds gradually increases with density [9]. This tendency has not been confirmed theoretically. Thus, the RTE model proposes that the effective reflectance is independent of the density [17]. However, comparison with ray tracing [16,21] shows that this model systematically underestimates the reflectance.

In summary, ray tracing appears to be the most reliable theoretical approach to radiative transfer in the powder beds but principal difficulties arise in modeling the packing structure. There is no yet a systematic theoretical study of the packing density influence in the range important for laser powder-bed applications. The present work proposes to analyze a set of regular structures of spheres within a wide range of packing density. The principal questions to answer concern the difference in radiative transfer between high-symmetry lattices of packed spheres and the influence of the structure orientation relative to the packed bed surface.

2. Ray tracing

The great particle size/wavelength ratio justifies using ray tracing. In a packed bed, the ray tracing is not applicable in thin gaps between particles around contact points. The thickness of such a gap is of the order of the wavelength. Therefore, the cross section area of the gap is much less than the projected area of a particle. Thus, the relative error introduced is expected to be low.

The example of Fig. 1 shows that one can distinguish incident rays subjected to one, two etc. reflections from a non-convex opaque object. A novel ray tracing technique is proposed. It is ap-

Table 1
Structure models of powder beds.

Structure	Abbreviation	Coordination number	Density
3D lattices of packed equal spheres			
Face-centred cubic	FCC	12	$\pi\sqrt{2}/6 = 0.740$
Hexagonal close-packed	HCP	12	$\pi\sqrt{2}/6 = 0.740$
Body-centered cubic	BCC	8	$\pi\sqrt{3}/8 = 0.680$
Simple cubic	SC	6	$\pi/6 = 0.524$
Diamond	DI	4	$\pi\sqrt{3}/16 = 0.340$
2D lattices of packed equal circles			
Hexagonal	Hex	6	$\pi\sqrt{3}/6 = 0.907$
Square	Sqr	4	$\pi/4 = 0.785$
Hexagonal honeycomb	HexH	3	$\pi\sqrt{3}/9 = 0.605$

plicable in assumption that reflectivity ρ is independent of the incidence angle. In this case, every reflection of a ray attenuates ray's energy by a factor of ρ . Then, the energy becomes ρ^n of the initial energy after n reflections. Every ray is traced until it is reflected back or passes forward through the object analyzed. Then, the following series give the effective reflectance R and transmittance T of the object:

$$\begin{pmatrix} R \\ T \end{pmatrix} = \sum_{n=0}^{\infty} \begin{pmatrix} r_n \\ t_n \end{pmatrix} \rho^n, \quad (1)$$

where r_n is the fraction of incident rays turned back after n reflections and t_n the fraction of incident rays passed forward after n reflections from the object. Here $r_0 = 0$ by definition because no ray can turn back without a reflection while t_0 can be positive. Of course, all the incident rays are directed on the object. However, the object may have through holes. Therefore, transmittance through the object is possible without reflections.

The assumption that ρ is independent of the incidence angle, is acceptable for metals [24]. The advantage of such an approach is that the ray tracing computation is independent of the value of reflectivity ρ . Once factors r_n and t_n are calculated by ray tracing, they can be applied to any combination material/wavelength defining the value of ρ . This offers the generality of results necessary for the present study. Numerical ray tracing computation is stopped if a ray is not reflected back or transmitted forward after a predefined great number of reflections N . Therefore, the series in Eq. (1) are truncated in numerical calculation. Partial sums of N terms approach R and T with the confidence interval estimated as

$$0 < \begin{pmatrix} R \\ T \end{pmatrix} - \sum_{n=0}^N \begin{pmatrix} r_n \\ t_n \end{pmatrix} \rho^n \leq E_N \rho^{N+1}, \quad (2)$$

where

$$E_N = 1 - \sum_{n=1}^N r_n - \sum_{n=0}^N t_n, \quad (3)$$

is the fraction of rays neither turned back nor passed forward after N reflections. The left inequality of Eq. (2) follows from the fact that all terms of the series in Eq. (1) are positive. In the right inequality of Eq. (2), fraction E_N given by Eq. (3) is the upper bound for both the factors r_{N+1} and t_{N+1} .

Table 1 lists the studied periodic structures with their principal characteristics, the coordination number and the relative density. Typical powder beds used in laser processing has the density in the range from 0.3 to 0.67 [9]. The density depends on the interplay of the gravity and the Van der Waals adhesive forces and varies with the powder material and particle size [21]. The regimes of mechanical spreading of powder also influence the resulting powder bed density [12]. The mentioned range of density is approximately covered by the DI, SC, and BCC structures, see Table 1. The close packed FCC and HCP structures are considerably more

dense. However, close packing is believed to be favorable to improve the laser powder bed fusion process and is considered in the ray tracing simulation of Boley et al. [16] and Zhang et al. [10]. The present study includes the FCC and HCP structures to compare the results with the previous works. Besides, one cannot exclude another application domain where the theoretical results on FCC and HCP would be demanded.

For each structure, surfaces parallel to several low-index planes are tested as shown in Figs. 2 and 3. These figures also show the calculation domains for ray tracing chosen for each combination structure/reflecting surface. The optimal calculation domain is a prism bounded by vertical planes in 3D (see Fig. 2) and a rectangle in 2D (see Fig. 3). Everywhere excluding DI(110), the nearest vertical mirror planes of the given structure are chosen as the lateral boundaries of the calculation domain. In the case of DI(110) shown in Fig. 2(l), one cannot find the necessary number of mirror planes perpendicular to the (110) plane. Therefore, two parallel mirror planes and two parallel planes perpendicular to them bound the calculation domain from the lateral sides. The distance between the no-mirror planes equals the lattice period in the direction normal to them.

Vertical incident rays are generated on the top of the calculation domain. The mirror conditions are imposed on the mirror lateral boundaries. The periodic conditions are imposed on the non-mirror lateral boundaries. The spheres reflect the rays specularly. A ray is traced until the number of reflections from the spheres attains a predefined number N or it exits through the top or bottom boundary of the calculation domain. Fractions r_n and t_n are calculated as the ratio of rays exited from the top and bottom boundaries, respectively, after n reflections from the spheres to the total number of incident rays. The number of incident rays for each calculation is of the order of 10^{10} . Appendix A presents analytical calculation of factors r_1 and r_2 for the close-packed row of circles in two dimensions (see Fig. 3(a) or (c)) that were used to test the algorithm for numerical calculations.

3. Normal-hemispherical reflectance and absorptance

Fig. 4 shows the first 12 reflection factors r_n (a) numerically calculated by ray tracing and the effective absorptance (b) of the 2D structures. It has to be noted that there are no through gaps in such structures. Therefore, transmittance $T = 0$ and absorptance $A = 1 - R$, where R is calculated by Eq. (1). The effective absorptance is plotted versus the absorptance of a flat surface $a = 1 - \rho$ in the range from 0.01 to 1. For the close-packed row of circles corresponding to Sqr(10) or Hex structures, the first two reflection factors calculated by ray-tracing (black circles in Fig. 4(a)) coincide with the corresponding values calculated analytically in Appendix A (crosses). Analytical calculations become very cumbersome for three and more reflections. Therefore, only the cases of one and two reflections are considered. The value of the reflection factor r_n depends on the kind of 2D structure, especially in

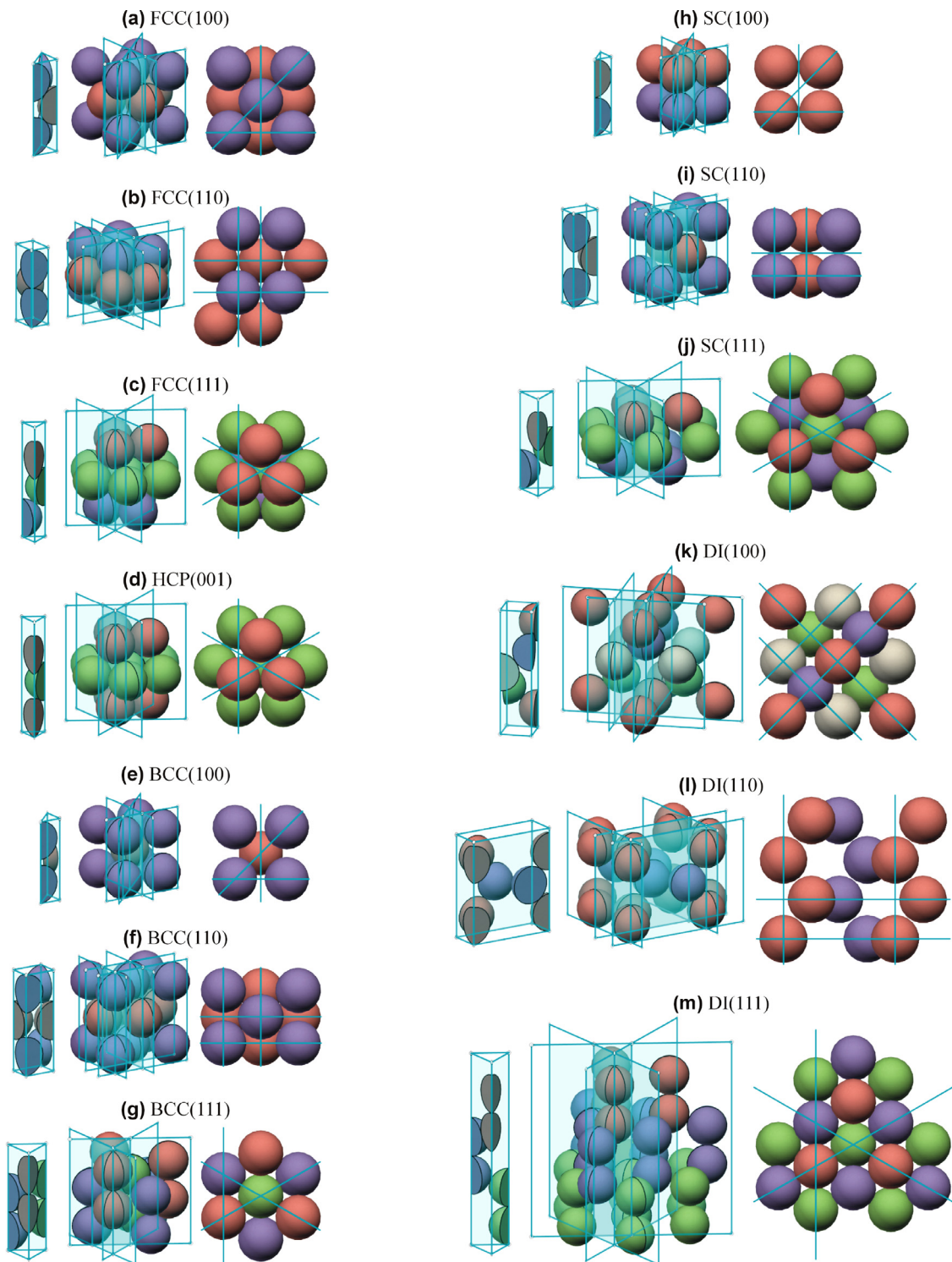


Fig. 2. Beds of equal spheres packed into regular structures with the top horizontal surface parallel to the crystallographic plane indicated by the Miller indices: (a)–(c) FCC; (d) HCP; (e)–(g); (h)–(j) SC; (k)–(m) DI. Three views are shown for each powder bed model: Top view on the reflecting surface (right); Axonometric view with the reflecting surface on the top (middle); Calculation domain for ray tracing (left).

the range of n from 3 to 10 (see Fig. 4(a)). Nevertheless, the effective absorptance is not sensitive to either the structure or the orientation (see Fig. 4(b)). Probably, it is because r_1 and r_2 are nearly identical for all the structures, which have the most weight in Eq. (1).

Fig. 5 compares the effective reflectance R for beds consisting of close-packed monolayers. The stacking is ABAB ... for the HCP

structure and ABCABC ... for the FCC one (see Fig. 2(c) and (d)). The first two layers are common for FCC(111) and HCP(001). Two layers reflect noticeably more than one layer (see Fig. 5(b)), especially for highly reflective materials with low a . It is in line with the reflection factors r_n (see Fig. 5(a)); the difference between them increases with n while the terms with great n are important for highly reflective materials only, according to Eq. (1). The difference

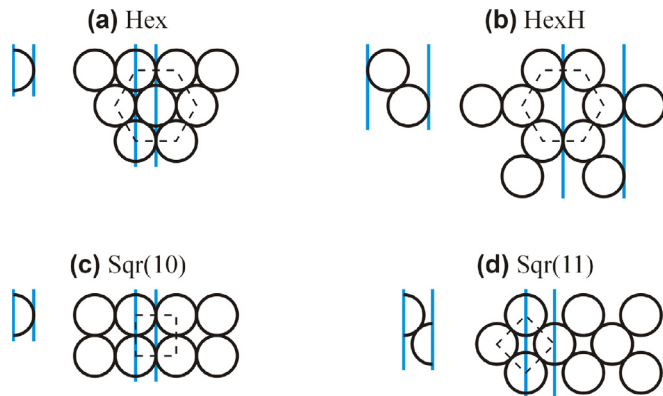


Fig. 3. Beds of equal circles packed into regular 2D structures: (a) Hexagonal lattice with a close-packed row of circles on the top; (b) Hexagonal honeycomb lattice; (c) Square lattice with a close-packed (10) row of circles on the top; (d) Square lattice with a (11) top plane; Two views are shown for each powder bed model: The lattice with an elementary cell (dashed lines) and two nearest vertical mirror lines (right); Calculation domain for ray tracing (left).

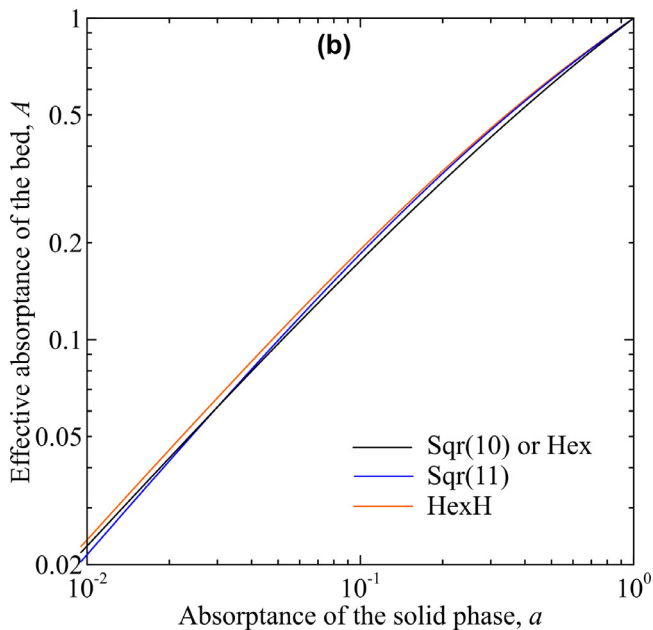
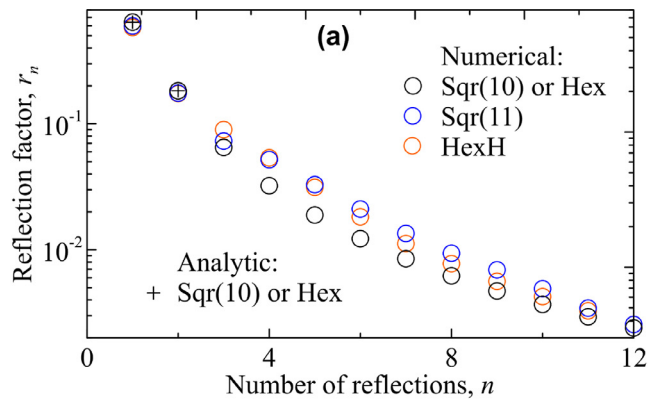


Fig. 4. Beds of packed circles in 2D: reflection factors (a) and effective absorbance (b).

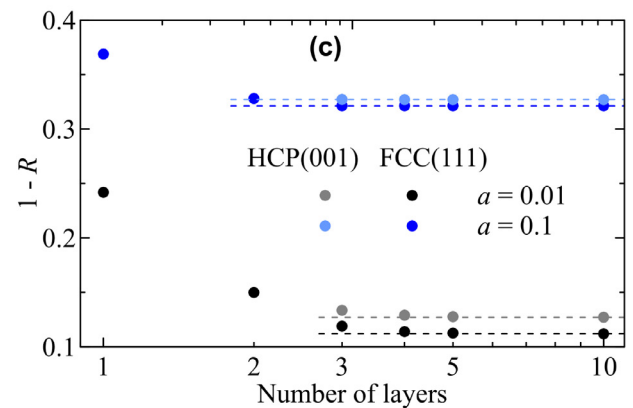
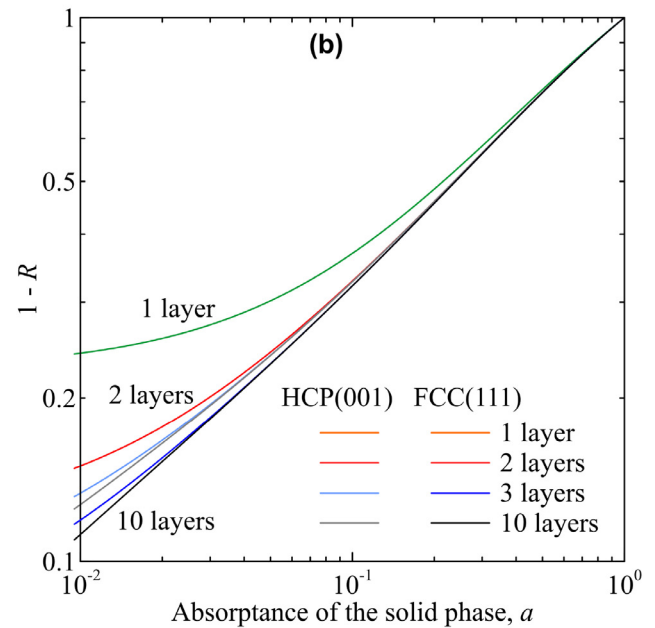
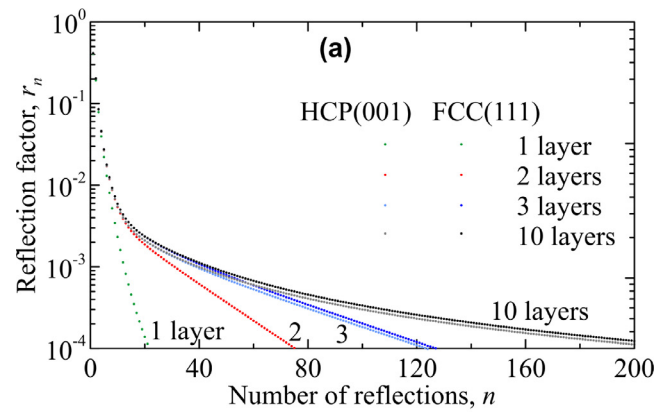


Fig. 5. Stacking of close-packed layers of spheres: reflection factors (a), effective reflectance R (b), and convergence of the effective reflectance with the number of layers (c).

between the beds of 2 and 3 close-packed layers is considerably less and can be neglected at $a > 0.1$. The three layers packed in FCC(111) are a bit more reflective than the three HCP(001) layers. That is because the third FCC(111) layer overlaps all the through holes while there are through holes in the HCP(001) stacking at any number of layers (compare the top views in Figs. 2(c) and (d)). However, the difference between FCC(111) and HCP(001) becomes

Table 2

Models of deep powder beds: Parameters of sphere structures and ray tracing. The sphere diameter is D .

Structure	Plane	λ/D	K	S/D	N	E_N
FCC	(100)	$\sqrt{2}/2$	10	7.07	400	0.012
	(110)	$1/2$	20	10	600	0.015
	(111)	$\sqrt{2}\sqrt{3}/3$	10	8.16	2000	$2.8 \cdot 10^{-3}$
HCP	(001)	$\sqrt{2}\sqrt{3}/3$	10	8.16	2000	$5.6 \cdot 10^{-4}$
BCC	(100)	$\sqrt{3}/3$	20	11.55	600	0.012
	(110)	$\sqrt{2}\sqrt{3}/3$	20	16.33	600	0.026
	(111)	$1/3$	30	10	700	$6.6 \cdot 10^{-3}$
SC	(100)	1	20	20	1000	$3.4 \cdot 10^{-3}$
	(110)	$\sqrt{2}/2$	20	14.14	800	$9.8 \cdot 10^{-4}$
	(111)	$\sqrt{3}/3$	20	11.55	400	$5.2 \cdot 10^{-3}$
DI	(100)	$\sqrt{3}/3$	40	23.09	500	$2.5 \cdot 10^{-3}$
	(110)	$\sqrt{2}\sqrt{3}/3$	20	16.33	500	$1.2 \cdot 10^{-4}$
	(111)	1, 1/3, 1...	30	20	600	0.022

negligible at $a > 0.1$. The reflection factors r_n along with the effective reflectance R converge rapidly with the increase of the number of the monolayers stacked. Fig. 5(c) shows that the reflectance of five layers is essentially the same as the reflectance of ten layers in the studied range of a , $0.01 \leq a \leq 1$. Thus, the curves of reflectance for 10 layers shown in Fig. 5(b) should be an accurate estimate for an infinitely deep powder bed composed of close-packed layers parallel to the surface. In summary, the top layer and the layer below the top one essentially form the back reflected radiation. The contribution of the third layer is much less important and the influence of the fourth and the consecutive layers can be neglected depending on the particular conditions and the required accuracy.

In the majority of experimental works, the reflectance is measured for deep powder beds. To obtain the ray tracing results corresponding to the experimental conditions, the influence of the bed thickness is numerically studied for all the cases shown in Fig. 2. The beds of different thicknesses are obtained by consecutive addition of the layers of spheres parallel to the surface. The convergence analysis is similar to those for the close-packed structures, which is described above in detail. The difference is that the layers are not close-packed. The interlayer distance depends on the given structure and the direction. Fig. 2 shows the spheres belonging to different layers by different colors. The convergence analysis reveals the number of layers K and the corresponding bed thickness S at which the effective reflectance becomes independent of the depth. The ray tracing results obtained with K layers closely correspond to the reflectance of a deep bed. Table 2 shows the number of layers K , the bed thickness S , the number of reflections N taken in to account at ray tracing, and the calculated value of parameter E_N necessary to estimate the error by Eq. (2). The convergence with respect to the number of testing incident rays has been studied by comparing the results calculated for approximately 10^8 , 10^9 , and 10^{10} rays. Even at 10^9 , the results are satisfactory but a bit noisy. At 10^{10} rays, the noise becomes non-visible in the plots. The principal calculations are accomplished for the number of testing incident rays around 10^{10} . For all the structures except for DI(111), $S = K\lambda$, where λ is the interlayer distance. For DI(111), the interlayer distances of 1 and $1/3$ sphere diameter D alternate, and $\lambda/D = 2K/3$ for even values of K .

Top views in Fig. 2 show that many of the considered regular structures of spheres has through holes. This means that the transmittance will not tend to zero with increasing the thickness. However, if particles are ordered in realistic powder beds, the characteristic size of ordered domains does not exceed several particle diameters. For example, discrete-element modeling of formation of a powder bed [12,20] revealed short-range ordering within 3–4 particle diameters. Thus, a deep powder bed should consist of small ordered domains. Radiation transmitted through an ordered

domain can be completely scattered and absorbed by the neighbor domains. Indeed, experiments [19] indicated that the transmittance of powder beds becomes not detectable at the thickness of about 5 particle diameters. Therefore, one can neglect the transmittance of a deep powder bed and estimate its absorptance as $A = 1 - R$. Fig. 6 shows the absorptance found. It has to be noted that the absorptance significantly depends not only on the structure but also on the orientation of the structure relative to the top surface of the bed, which is specified by the Miller indices of the top layer of spheres. The dependence on orientation is especially important for the less dense SC and DI structures.

The reflectance experiments average the radiation reflected from a great number of the ordered domains adjacent to the powder bed surface. Suppose that the different domains are randomly oriented. Then, the ray-tracing results can be averaged over orientation to approach the experimental conditions. Fig. 7 shows that there are three (100), four (111), and six (110) directions in cubic lattices. Therefore, the weighted average absorptance is

$$A = \frac{3}{13}A_{100} + \frac{6}{13}A_{110} + \frac{4}{13}A_{111}, \quad (4)$$

where indices designate orientation. Fig. 8 summarizes the absorptance averaged over orientation. It has to be noted that all the structures of packed spheres are considerably more absorptive than 2D structures of packed circles, for example Sqr(10) (see Fig. 8), and the effective absorptance successively increases in the series of FCC, BCC, SC, and DI structures. Table 1 indicates that the packing density decreases in this series. Thus, the ray tracing reveals a considerable increase of the effective absorptance with decreasing the density.

Gusarov and Kruth [17] approximately calculated the absorptance of a deep powder bed in the framework of the model of equivalent absorbing scattering medium. A two-moment approach to the radiative transfer equation resulted in the following expressions for effective reflectance R_{em} and absorptance A_{em} [17]:

$$R_{em} = \frac{1 - \sqrt{a}}{1 + 2\sqrt{a}}, \quad A_{em} = 1 - R_{em} = \frac{3\sqrt{a}}{1 + 2\sqrt{a}}. \quad (5)$$

The dash-dotted line in Fig. 8 shows A_{em} . The ray tracing results appear to approach A_{em} when the packed density of the bed tends to zero. The greater the density of the bed, the greater the deviation of the ray tracing from the equivalent-medium theoretical prediction. The equivalent-medium model [17] assumes that the incident radiation enters into the spaces between the particles and neglects the radiation projected directly on the top particles. This can explain the discrepancy observed. Introducing the appropriate boundary conditions for the incident radiation one can correct the drawback of the equivalent-medium model. Indeed, fraction r_1 of the incident rays is reflected back by the particles on the top of the powder bed according to the approach applied in the present work. This fraction of the incident radiation does not penetrate into the powder bed. It is natural to exclude it from the equivalent-medium model, which gives

$$R = r_1\rho + (1 - r_1)R_{em}, \quad A = 1 - R = r_1a + (1 - r_1)A_{em}. \quad (6)$$

One can consider Eq. (6) as the first-term approximation of Eq. (1) with the remainder estimated by the equivalent-medium model.

Rigorous application of Eq. (6) requires numerical calculation of the first reflection factor r_1 by ray tracing. Fig. 9(a) shows a spherical particle on the top of a powder bed and explains how to estimate the value of r_1 without numerical ray tracing. If a normally incident ray strikes the sphere at a point of the 45-degrees cone shown in Fig. 9, it is reflected parallel to the bed surface. The sphere reflects back any ray striking the spherical segment cut by the cone. The projected area of this segment S_b is a half the pro-

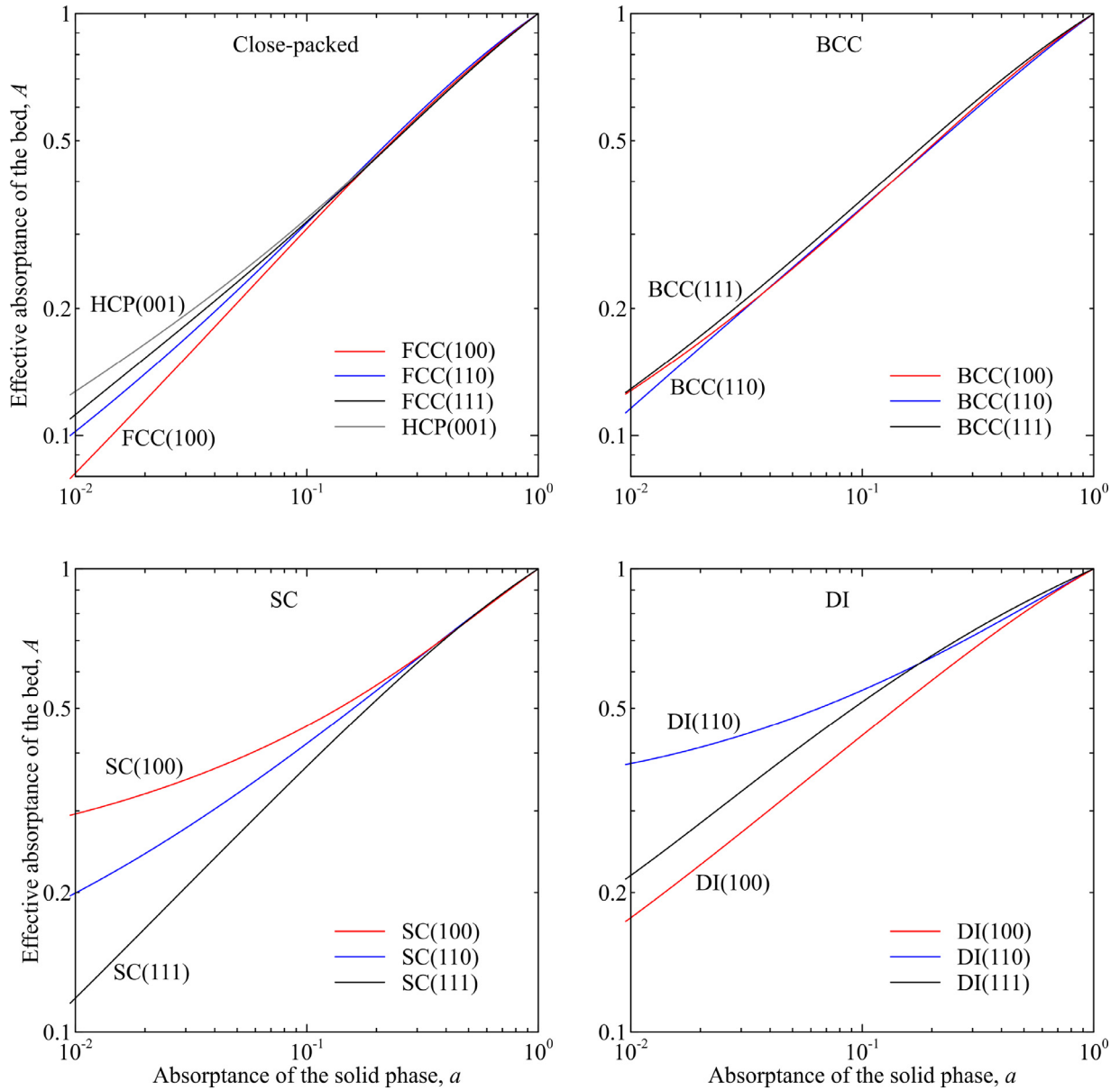


Fig. 6. Effective absorptance of a deep bed of equal spheres depending on the structure and the top surface indicated by the Miller indices.

jected area of the sphere S ,

$$S_b = S/2. \quad (7)$$

Suppose that there are k spherical particles on the top of the powder bed with the total area S_{tot} . Then, the packing density is

$$d = \frac{kS}{S_{tot}}, \quad (8)$$

according to the principle of Delesse. The first reflection factor is

$$r_1 = \frac{kS_b}{S_{tot}}. \quad (9)$$

One can obtain from Eqs. (7)–(9) that r_1 is a half d ,

$$r_1 = \frac{d}{2}. \quad (10)$$

Fig. 9(b) shows that Eq. (10) satisfactorily estimates the numerically calculated factors r_1 . Substitution of Eq. (10) into Eq. (6) results in the following analytic approximation for the effective ab-

sorptance:

$$A = \frac{d}{2}a + \left(1 - \frac{d}{2}\right) \frac{3\sqrt{a}}{1 + 2\sqrt{a}}. \quad (11)$$

Fig. 8 shows this approximation by dashed lines. The modified equivalent medium model gives the same tendency as the ray tracing calculation (full lines). The insert in Fig. 8 shows the relative deviation between the model and the ray tracing estimating the error of the model. The relative error for SC is less than 10%. The relative error for DI becomes greater than 10% only at low absorptance of the solid phase $a < 0.03$. The approximation of BCC and FCC is worse. For these structures, the relative error increases above 15% at $a < 0.2$. Thus, the precision of the modified model of equivalent medium decreases with the packing density in the series of DI, SC, BCC, and FCC structures.

Fig. 8 shows experimental data on absorptance of powder beds reported by Boley et al. [11], Tolochko et al. [8], and Gusarov et al. [9]. The packing density d of the studied spherical powders is in the range from 0.5 to 0.6 [9], which approximately corresponds to the SC structure with $d = 0.524$. Indeed, most of the experimental

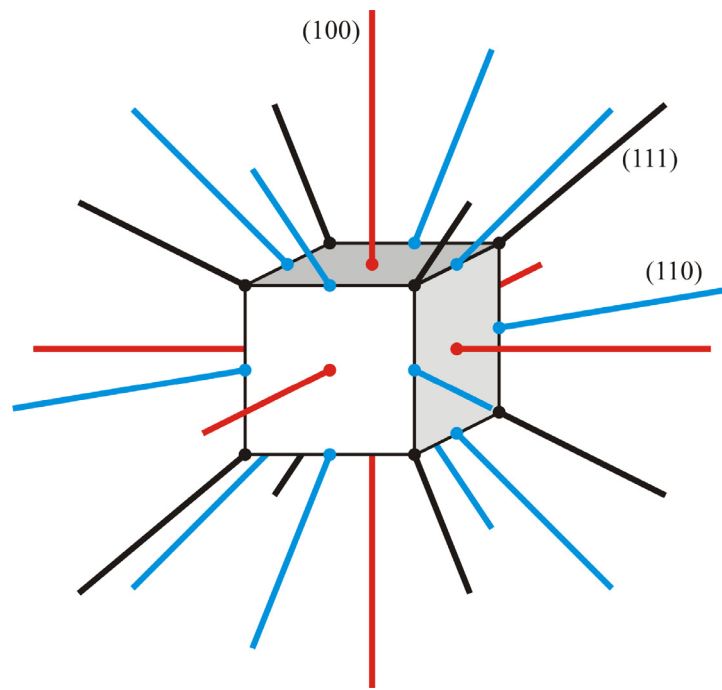


Fig. 7. Low-index directions in a cubic lattice.

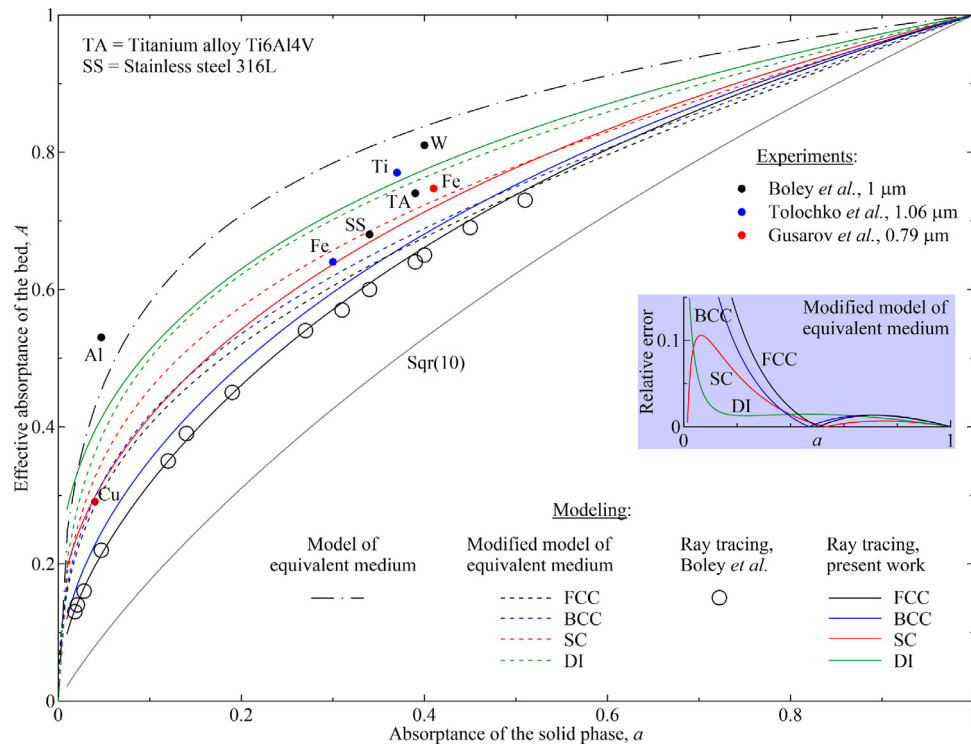


Fig. 8. Average effective absorptance of a deep bed of equal spheres compared with the absorptance of Sqr(10) structure in 2D, other models, and experiments. The insert estimates the relative error of the modified model of equivalent medium.

points are near the full red curve calculated by ray tracing and the dashed red curve calculated by the modified equivalent-medium model, Eq. (11), for the SC structure. However, the powders of Al, Ti, and W are significantly more absorptive and correspond better to the equivalent medium model, Eq. (5). It would not validate the equivalent medium model because it contradicts the recent ray-tracing results. Boley *et al.* [11] supposed that aluminum is oxidized and, therefore, absorbs more. Titanium and tungsten

are easily oxidized too. The oxidation can explain the discrepancy between the experiments and the models for these three powders. Gusarov *et al.* [9] measured the effective reflectance of an oxidized copper powder and the same powder after annealing in hydrogen atmosphere. They found that the annealed powder is considerably less absorbing. The red point for copper in Fig. 8 corresponds to the annealed powder. It perfectly agrees with the ray tracing (see the red full-line curve).

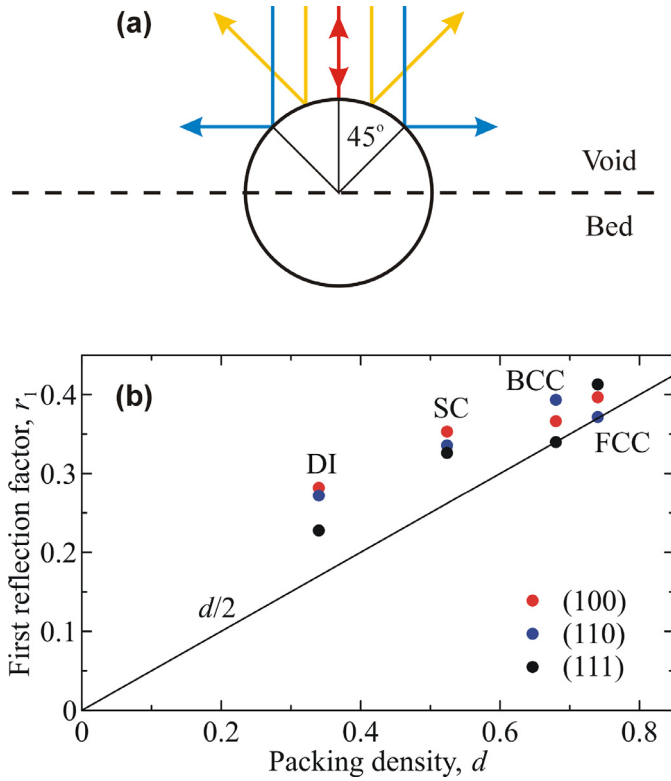


Fig. 9. (a) Back reflection by a sphere on the top of a powder bed. (b) First reflection factor r_1 calculated by ray tracing (points) and theoretically estimated (line).

The same metal can follow the specular or diffuse reflection model depending on surface roughness, oxidation degree, and other factors influencing the state of the surface. It is difficult to detect experimentally if the surface of a given powder particle is specular or diffuse because of the small size, typically below 50 μm . Powders obtained by gas-phase atomization frequently have spherical particles with a rather smooth surface, see SEM images in Ref. [9]. One can expect that they reflect specularly. Indeed, oxidation can change the reflection model. We tried to estimate the difference between the specular and diffuse reflection models theoretically [17]. It has to be noted that the reflection pattern of a single convex opaque particle irradiated by a collimated beam is isotropic for a specular surface and anisotropic, preferentially backward directed, for a diffuse surface [17]. This explains the modeling results [17] indicating that the bed of diffuse particles is up to 10% more reflective. The experimental trend is opposite: oxidation makes powder beds more absorptive. Therefore, the change of reflection model at oxidation cannot be the principal factor. It is likely that the oxidized surface becomes more absorptive.

Experiments of Gusarov et al. [9] revealed a dependence of the powder bed absorptance on the density of the bed. Closed circles in Fig. 10 show these data for iron and copper powders. The confidence interval reported is ± 0.002 [9]. It is much smaller than the symbols in Fig. 10. The figure indicates a clear decrease of the absorptance with the density for both of studied materials. Open circles in Fig. 10 show the ray tracing results for these materials. The absorptance of solid material a accepted for modeling is equal to the value estimated from the complex refractive index [9]. The values of a are shown on the right-bottom of Fig. 10. The ray tracing gives the similar function $A(d)$. A small shift of the ray tracing relative to the experiments for Fe can be explained by an error in the accepted value of a . Full lines in Fig. 10 show the modified equivalent-medium model, Eq. (11), for the same values of a . The curve for Fe satisfactorily agrees with both the experiments and

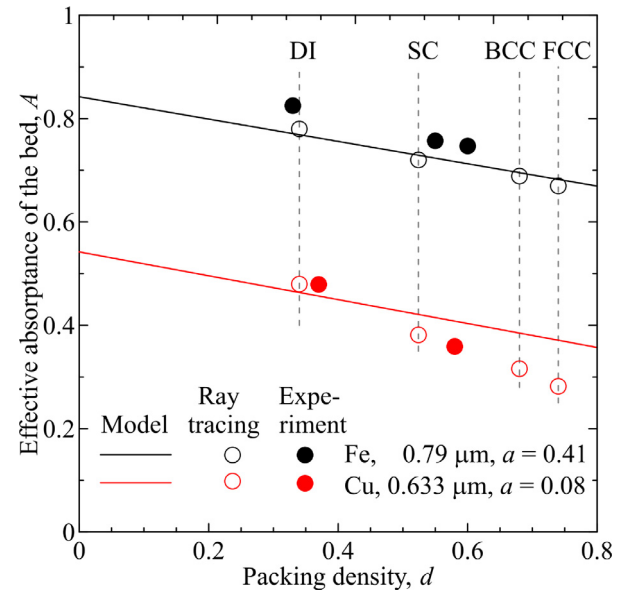


Fig. 10. Powder bed absorptance versus the packing density: Comparison of experiments of Gusarov et al. (closed circles) with the ray tracing (open circles) and the modified equivalent-medium model (full lines).

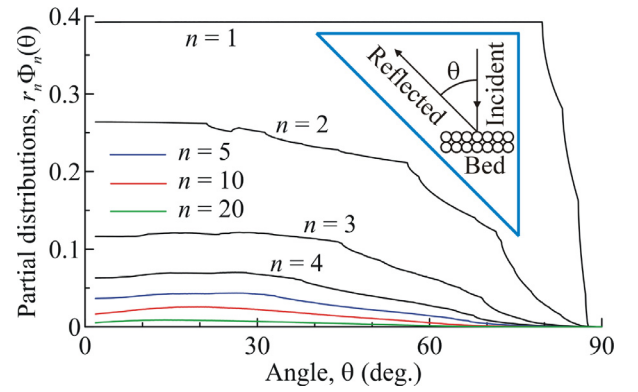


Fig. 11. Partial angular distributions of reflected radiation $\Phi_n(\theta)$ for 20 layers of SC(100) structure. The insert defines the reflection angle θ .

the ray tracing. The curve for Cu demonstrates the right tendency of A decreasing with d . It approximates the experiment and the ray tracing at $d < 0.4$ but a discrepancy arises and increases at $d > 0.4$.

4. Normal-directional reflectance

The direction of the reflected radiation is commonly characterized by polar θ and azimuth φ angles relative to the external normal to the powder bed surface. The studied regular structures exhibit considerable variation of the reflected distribution both in θ and φ . However, the variation in φ does not correspond to realistic powder beds where all in-plane directions are statistically equivalent. According to the reasoning reported in the previous section, the statistically isotropic packed bed is likely a set of randomly oriented small ordered domains. This suggests averaging the angular distribution over φ . Only distributions in polar angle are considered below. In the chosen frame, the polar angle is equivalent to the reflection angle.

To estimate the angular distribution of reflected radiation, reflection angle θ is calculated for every back-reflected ray during ray tracing. The insert in Fig. 11 shows the reflection angle between the reflected ray and the external normal to the powder bed surface. The interval of angles from 0 to $\pi/2$ is divided into a great

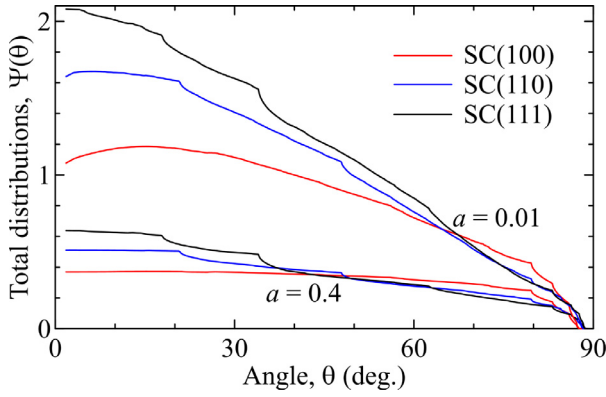


Fig. 12. Total angular distributions of reflected radiation $\Psi(\theta)$ for the models of a deep powder bed with the SC structure at selected values of the solid-phase absorptance a .

number of small intervals. The number of rays within an interval around angle θ is divided by the solid angle covered by this interval, which gives the value of the angular distribution at θ . We distinguish angular distributions $\Phi_n(\theta)$ of rays back-reflected after n reflections from spheres and normalize them,

$$\int_0^1 \Phi_n(\theta) d \cos(\theta) = 1. \quad (12)$$

Then, the total angular distribution of reflected radiation is

$$\Psi(\theta) = \sum_{n=1}^{\infty} r_n \rho^n \Phi_n(\theta). \quad (13)$$

Series (13) is similar to Eq. (1) for the normal-hemispherical reflectance R . Integration of Eq. (13) gives the normalization condition,

$$\int_0^1 \Psi(\theta) d \cos(\theta) = R. \quad (14)$$

Fig. 11 shows selected partial angular distributions $\Phi_n(\theta)$ used as basis functions in series (13).

Fig. 12 shows examples of calculation by Eq. (13) for the models of a deep powder bed with the SC structures of various orientations specified by the Miller indices of the top plane at the selected absorptance values of a flat surface of the solid phase $a = 1 - \rho$. The number of layers of spheres K and the number of terms in the expansion N are listed in Table 2. The results show that the angular distribution of the reflected radiation strongly depends on the structure orientation relative to the top surface of the powder bed. Therefore, the weighted average is calculated according to the frequency of the three low-index crystallographic directions shown in Fig. 7. The averaging is similar to Eq. (4),

$$\Psi(\theta) = \frac{3}{13} \Psi_{100}(\theta) + \frac{6}{13} \Psi_{110}(\theta) + \frac{4}{13} \Psi_{111}(\theta), \quad (15)$$

where indices designate orientation. Fig. 13 compares the angular distributions averaged over orientation for the sequence of structures FCC, BCC, SC, and DI with decreasing packing density d . All the distributions are cosine-like ones typical for diffuse reflection. One can select features typical for each structure, for example, a small peak at $\theta = 0$ obtained for the FCC. However, there is no general tendency of varying the angular distribution with the packing density. For all the studied structures, the angular distribution deforms with increasing the solid-phase absorptance a : the relative contribution of low-angle ($\theta < 50^\circ$) reflections decreases and that of high-angle ($60^\circ < \theta < 85^\circ$) reflections increases (see Fig. 13).

Fig. 14 compares the ray tracing with experimental data [9] on the normal-directional reflectance defined as the ratio of the angular distribution of the reflected radiation to that of totally reflecting

diffuse surface,

$$R(\theta) = \frac{\Psi(\theta)}{2 \cos(\theta)}. \quad (16)$$

The powders of iron and copper consisted of spherical particles with the diameter approximately from 125 to 160 μm . Their packing density d was 0.55 and 0.58, respectively [9]. It is around the packing density of the SC structure, see Table 1. That is why this structure is chosen for ray tracing. The values of the solid-phase absorptance a indicated in Fig. 14 are taken as reported in Ref. [9]. The experiment and the modeling excellently agree at reflection angles $\theta < 70^\circ$ for Cu and $\theta < 80^\circ$ for Fe. The modeling confirms the experimentally observed tendency of increasing the normal-hemispherical reflectance with θ at $\theta > 60^\circ$. A discrepancy at oblique angles can be explained by the experimental errors due to the influence of the borders of the cuvette with the powder and a non-ideal alignment of the sensor relative to the cuvette [9].

5. Penetration of radiation into the powder bed

The depth profile of the radiative energy flux is calculated during the ray tracing. Axis Z is directed downward, codirected with the incident rays. The centers of the top-layer spheres are placed at the level $z = 0$. A great number of horizontal planes $z = \text{const}$ are uniformly distributed over the height of the calculation domain. The number of intersections of rays with every plane is counted. We count the rays with positive and negative projections on axis Z separately to calculate the net flux of rays at depth z as the difference of the intersections of rays with the positive projections $F^+(z)$ and those of rays with the negative projection $F^-(z)$. The rays after n reflections by spheres are counted separately to obtain partial fluxes of particles $F_n^+(z)$ and $F_n^-(z)$. Finally, normalized partial net fluxes are calculated as

$$f_n(z) = \frac{F_n^+(z) - F_n^-(z)}{F_i}, \quad (17)$$

where F_i is the number of incident rays. Thus, $f_n(z)$ is the net fraction of incident rays intersecting level z in the positive direction of axis Z after n reflections by spheres. The energy transferred by a ray is multiplied by a factor of ρ^n after n reflections. Therefore, the ratio $q(z)$ of the net radiative energy flux at depth z to the energy flux transferred by the incident radiation can be expanded in series,

$$q(z) = \sum_{n=0}^{\infty} \rho^n f_n(z), \quad (18)$$

similar to Eq. (1) for the normal-hemispherical reflectance and Eq. (13) for the angular distribution of the reflected radiation.

Fig. 15 shows an example of the calculated partial net flux distributions. The depth z is normalized by particle radius $D/2$ here. The distributions are uniform above the powder bed at $z < -D/2$ and decay with z inside the powder bed. The flux of the incident rays f_0 tends to a non-zero value as z increases. This value is the fraction of rays propagating in the through holes of the structure. Fig. 16 shows examples of calculation by Eq. (18) for the models of a deep powder bed with the SC structures of various orientations specified by the Miller indices of the top plane at the selected absorptance values of a flat surface of the solid phase $a = 1 - \rho$. The number of layers of spheres K and the number of terms in the expansion N are listed in Table 2. Fig. 16 indicates that the depth profile of the energy flux is sensitive to the structure orientation relative to the top surface of the powder bed. Therefore, the weighted average is calculated according to the frequency of the three low-index crystallographic directions shown in Fig. 7. The averaging is similar to Eqs. (4) and (15),

$$q(z) = \frac{3}{13} q_{100}(z) + \frac{6}{13} q_{110}(z) + \frac{4}{13} q_{111}(z), \quad (19)$$

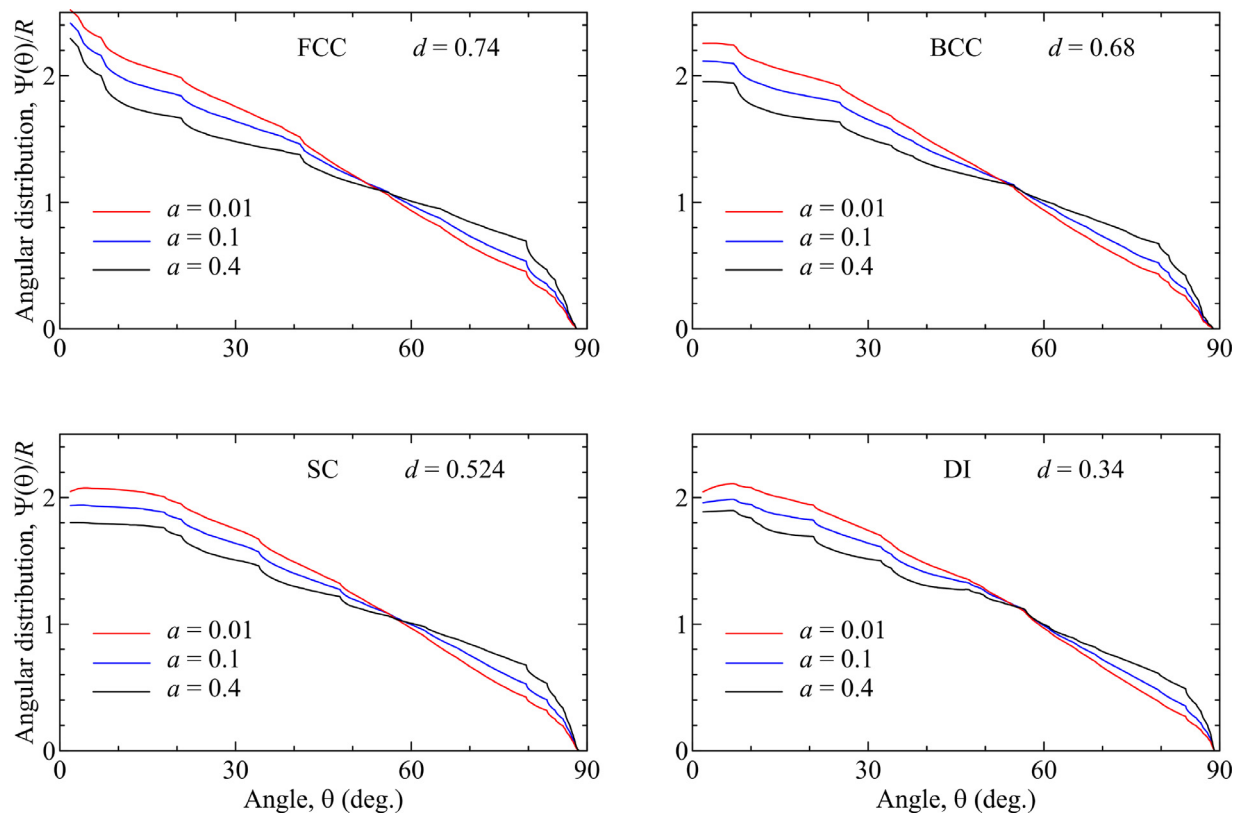


Fig. 13. Angular distribution of radiation reflected by a deep bed of equal spheres packed in the FCC, BCC, SC, and DI structures at selected values of the solid-phase absorptance a . Packing density d is indicated on the top of each diagram.

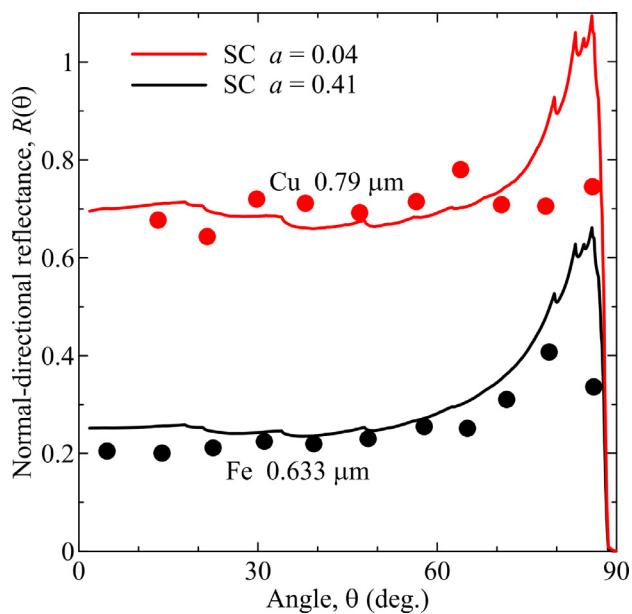


Fig. 14. Normal-directional reflectance $R(\theta)$: Comparison between the experiments of Gusarov et al. at 0.79 and 0.633 μm wavelengths (points) and the present ray tracing for the SC structure at the indicated values of the solid-phase absorptance a (lines).

where indices designate orientation.

The curves plotted in Figs. 12, 13, and 16 exhibit angular points, *i. e.* points where the derivative is not continuous. Probably, the angular points are due to shadowing of a particle by another one. It occurs at equivalent positions and directions in regular structures.

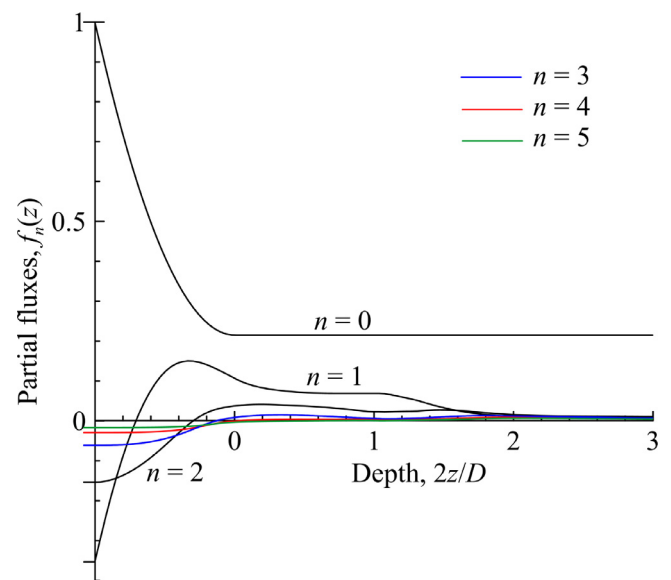


Fig. 15. Partial depth profiles $f_n(z)$ for 20 layers of SC(100) structure.

That is why averaging over a great number of particles does not eliminate the angular points. Such details cannot be generalized to random packing.

Full lines in Fig. 17 present the profiles of radiative energy flux averaged over orientation for the FCC, BCC, SC, and DI structures at various values of the flat-surface reflectance of the solid phase a . It has to be noted that the value of the normalized energy flux q at $z \leq -D/2$ is equal to the absorptance. The flux decays with z but generally does not tend to zero because a fraction of the in-

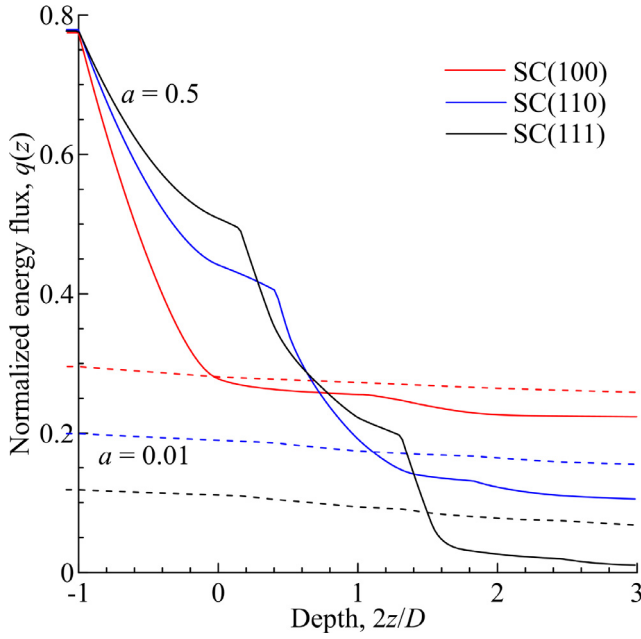


Fig. 16. Normalized net flux of radiative energy $q(z)$ for the models of a deep powder bed with the SC structure at selected values of the solid-phase absorptance a .

incident rays propagate in the through holes of the considered regular structures without reflections. In a realistic powder bed, the length of such through holes is limited by the size of quasi-regular domains and may be as low as few particle diameters [19]. Therefore, the penetration depth of the incident radiation is estimated as the characteristic length where the profile $q(z)$ changes from the constant left value to the asymptotic right value. Fig. 17 shows that the penetration depth can vary from few particle diameters in the more dense FCC and BCC structures of well-absorbing material with $a = 0.5$ to few tens of particle diameters in the less dense SC and DI structures of highly reflecting material with $a = 0.01$.

The radiation penetration depth determines the domain where the radiative energy transforms into the thermal one and thus considerably influences energy and heat transfer. Therefore, it is important for applications. Theoretical estimates are useful to predict such a highly variable value as function of material and powder bed parameters. The model of equivalent medium occupying half space $z > 0$ gives the following equation for the normalized flux of radiative energy [17]:

$$q_{em}(z) = \frac{3(\sqrt{a} - a)}{1 - 4a} \exp(-2\sqrt{a}\beta z) - \frac{3a}{1 - 4a} \exp(-\beta z), \quad (20)$$

where β is the effective extinction coefficient. The multiphase approach [25] estimates this parameter for the bed of opaque equal spheres as

$$\beta = \frac{3}{2} \frac{d}{(1 - d)D}. \quad (21)$$

The model of equivalent medium overestimates the absorptance of powder beds relative to the ray tracing results, see Fig. 8. The correction of the boundary condition for the half-space problem of radiation transfer is proposed in Section 3. This correction results in the following expression for the energy flux profile:

$$q(z) = \begin{cases} A = r_1 a + (1 - r_1)A_{em} & , \quad z \leq 0 \\ (1 - r_1)q_{em} & , \quad z > 0 \end{cases} \quad (22)$$

The estimation of the first reflection factor r_1 by Eq. (10) results in the analytic expression for the energy flux inside the powder bed

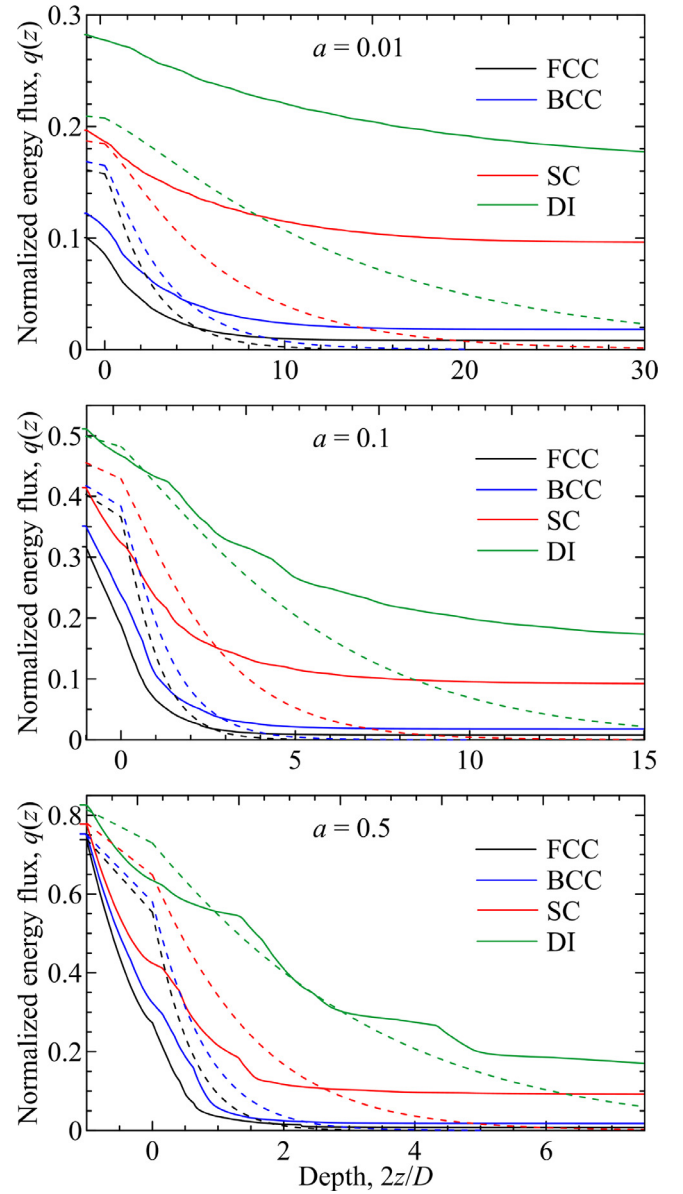


Fig. 17. Depth profile of radiative energy flux in a deep bed of equal spheres packed in the FCC, BCC, SC, and DI structures at selected values of the solid-phase absorptance a : ray tracing (full lines) and the modified model of equivalent medium (dashed lines).

at $z > 0$,

$$q(z) = \frac{3}{2} \frac{2 - d}{1 - 4a} [(\sqrt{a} - a) \exp(-2\sqrt{a}\beta z) - a \exp(-\beta z)]. \quad (23)$$

Dashed lines in Fig. 17 show the profiles of radiative energy flux according to the modified model of equivalent medium, Eq. (23). The domain between the sphere vertices at $z = -D/2$ and the centers of the top-layer spheres at $z = 0$ is considered as the intermediate layer between void, $z < -D/2$, and powder bed, $z > 0$. Therefore, the jump specified by Eq. (22) is spread over this domain by a linear approximation between $q(-D/2) = A$ and $q(0)$.

Fig. 17 shows that the modified model of equivalent medium (dashed lines) generally describes the considerable increase of the radiation penetration depth with decreasing both the packing density and the solid-phase absorptance revealed by the ray tracing (full lines). However, a significant difference may occur between the model and the ray tracing. To quantify the penetration of external radiation into a powder bed, we define the penetration depth

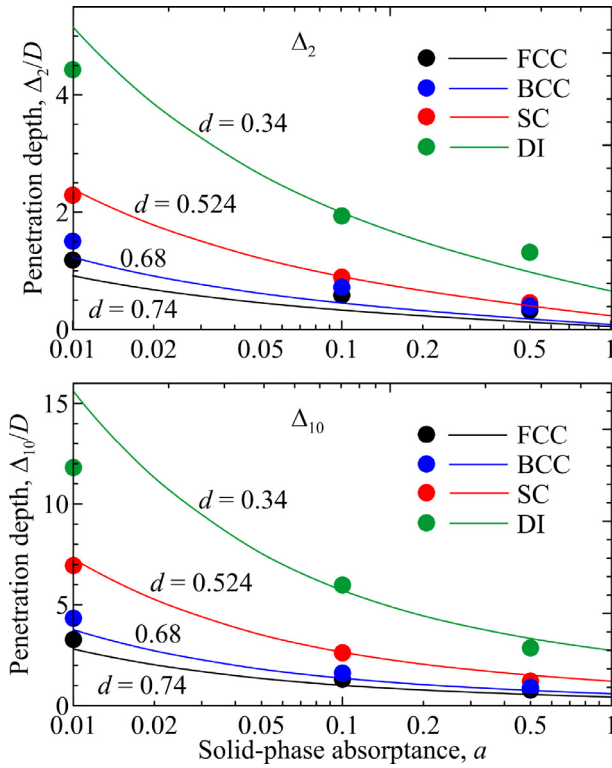


Fig. 18. Penetration depths of half-decay Δ_2 (top) and 90%-decay Δ_{10} (bottom) in a deep bed of equal spheres packed in the FCC, BCC, SC, and DI structures calculated by ray tracing (points) and the modified model of equivalent medium (lines). The packing density d is indicated near the curves.

Δ_k as the distance where the energy flux profile $q(z)$ decays from its maximum value $q(-D/2)$ down to $1/k$ fraction of the maximum above the asymptotic value $q(\infty)$. Namely,

$$\Delta_k = z_k + D/2, \quad (24)$$

where z_k is the solution of equation,

$$q(z) = q(\infty) + \frac{q(-D/2) - q(\infty)}{k}. \quad (25)$$

Fig. 18 compares the penetration depths at $k = 2$ and 10 calculated from the profiles of ray tracing and those of the model. The model satisfactorily agrees with the ray tracing for the depth of 90%-decay Δ_{10} in the range of the solid-phase absorptance $0.1 \leq a \leq 0.5$. Under other conditions, the model penetration depth may overestimate the ray-tracing depth but the model is still acceptable for order-of-magnitude estimates.

6. Conclusions

A ray tracing method is developed to simulate optical characteristics of heterogeneous media consisting of one transparent phase and one opaque phase. The principal assumption is that the reflectance of the opaque phase surface is a constant. The effective optical characteristics of the heterogeneous media are expanded in power series of the opaque-phase reflectance. Once the coefficients of the power series are calculated, they can be used for any combination wavelength/material thus facilitating the parametric analysis of the ray tracing results.

The ray tracing is applied to metal powder beds typical for laser powder-bed processing that are modeled by equal spheres packed in regular cubic-symmetry structures of various density, the face-centered cubic, hexagonal close-packed, body-centered cubic, simple cubic (SC), and diamond-like structures. The square, hexagonal,

and hexagonal honeycomb two-dimensional structures of equal circles are tested too. The simulation results are sensitive to the orientation of the studied structures relative to the surface of the powder bed. A realistic powder bed is assumed to consist of many small quasi-regular domains with arbitrary orientation. Therefore, the ray-tracing results are averaged over orientation.

To evaluate the penetration of external radiation into powder beds, radiative energy flux is calculated through a plane parallel to the bed surface at arbitrary depth z . The obtained profiles of the energy flux versus z revealed the dependence of the radiation penetration depth versus the packing density d and the solid-phase absorptance a . It can vary from few particle diameters for close-packed structures with $d = 0.74$ and moderate absorptance around $a = 0.5$ to few tens of particle diameters for a less dense powder bed of highly reflecting material with $d = 0.34$ and $a = 0.01$.

The absorptance at normal incidence and the normal-directional reflectance are calculated for deep powder beds. It is shown that the effective absorptance of a powder bed A considerably increases with decreasing the packing density d . The functions $A(d)$ obtained by the ray tracing agree with the known experimental data. The function $A(a)$ obtained by the ray tracing for the SC structure with $d = 0.524$ agrees with the known experimental data for powder beds of non-oxidized metallic spherical particles because the density of such powders approximately corresponds to that of the SC structure. The ray tracing can considerably underestimate the powder bed absorptance of Al, Cu, Ti, and W. It is explained by a possible oxidation of such powders, which can considerably decrease the reflectance of the solid phase.

The calculated angular distribution of the radiation reflected by a powder bed is similar to the cosine-like distribution corresponding to the diffuse reflection law. However, a deviation from the diffuse law is found: the reflectance in the directions around the normal is lower while the reflectance in oblique directions is greater. The deviation increases with the solid-phase absorptance a . The increase of the reflectance with the reflection angle agrees with the known experiments on the normal-directional reflectance of powder beds. The ray tracing does not reveal a general tendency for varying the normal-directional reflectance with the packing density d .

A modified model of equivalent medium is proposed to approximate the obtained ray tracing results by analytic expressions. The model uses the known solution to the half-space problem for the radiative transfer equation. The effective properties of the medium are estimated by the multiphase approach. The solution is modified by the proposed boundary condition to take into account that a fraction of the incident radiation is reflected back by the particles on the top of the powder bed without penetrating inside the bed. The precision of the modified model of equivalent medium decreases with the packing density. The model agrees with the ray-tracing predictions of the radiation penetration depth and agrees with the known experimental data on the powder-bed absorptance and normal-directional reflectance.

The radiative-transfer characteristics of powder beds calculated by ray tracing are validated by the available experimental data. They determine energy transfer in laser processing of powder beds. Therefore, the obtained results can be used for optimizing the existing laser powder-bed technologies and developing new ones.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

A.V. Gusarov: Conceptualization, Methodology, Software, Formal analysis, Writing - original draft.

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Appendix A. Analytic calculation of factors r_1 and r_2 in 2D

Fig. A1 traces rays striking a close-packed row of circles at the incidence normal to the row axis. The close-packed row is the top layer of both the Hex and the Sqr(10) strictures (see Fig. 3). Angle α in Fig. A1 specifies the position of the incident ray relative to the nearest circle. One can see that the incident ray turns back after the first reflection by the nearest circle at small α . Fig. A1(a)

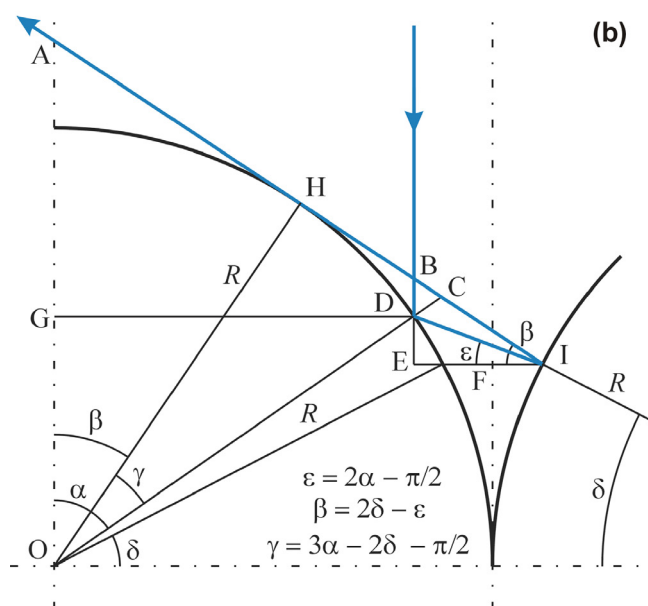
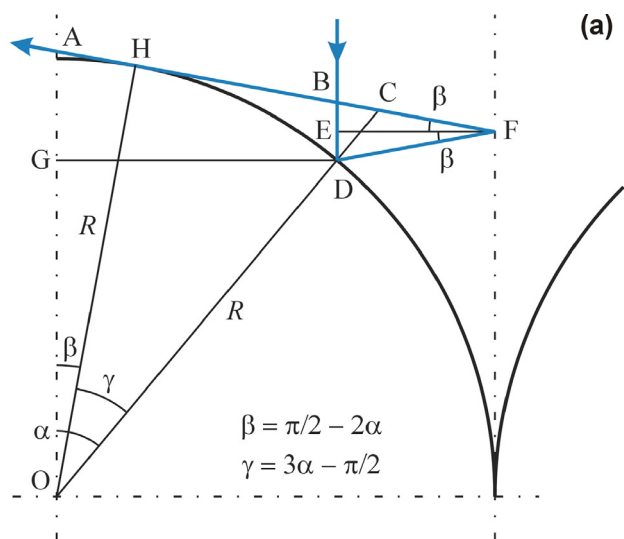


Fig. A1. Normal incidence on a close-packed row of circles: The upper bound of angle α when a ray turns back after the first reflection (a) and the second reflection (b).

shows the minimum angle $\alpha = \alpha_1$ when the ray is directed on another circle after the first reflection. We draw the ray reflected by a symmetry line in point F to make the figure more compact. Fig. A1(b) shows the minimum angle $\alpha = \alpha_2$ when the ray turns back after three reflections by the spheres. Thus, the incident ray turns back after one reflection if $0 \leq \alpha < \alpha_1$ and after two reflections if $\alpha_1 \leq \alpha < \alpha_2$. Therefore,

$$r_1 = \sin \alpha_1, \quad r_2 = \sin \alpha_2 - \sin \alpha_1. \quad (\text{A.1})$$

To find α_1 , consider similar triangles ACO and BCD in Fig. A1(a),

$$\frac{|AO|}{|BD|} = \frac{|CO|}{|CD|}. \quad (\text{A.2})$$

One can see from Fig. A1(a) that

$$|\text{CO}| - |\text{CD}| = R. \quad (\text{A.3})$$

From triangle AHO,

$$|\text{AO}| \cos \beta = R. \quad (\text{A.4})$$

From triangle CHO,

$$|\text{CO}| \cos \gamma = R. \quad (\text{A.5})$$

From Fig. A1(a),

$$|\text{DG}| + |\text{EF}| = R. \quad (\text{A.6})$$

From triangle DGO,

$$|\mathrm{DG}| = R \sin \alpha. \quad (\mathrm{A}.7)$$

From triangles BEF or DEF,

$$\frac{|\text{BD}|}{2} = |\text{EF}| \tan \beta. \quad (\text{A.8})$$

Exclude $|\text{DG}|$ and $|\text{EF}|$ from Eqs. (A.6)–(A.8):

$$|\text{BD}| = 2R(1 - \sin \alpha) \tan \beta. \quad (\text{A.9})$$

Exclude $|AO|$ and $|BD|$ from Eqs. (A.2), (A.4), (A.9):

$$|\text{CD}| = 2(1 - \sin \alpha) \sin \beta |\text{CO}|. \quad (\text{A.10})$$

Exclude $|CO|$ and $|CD|$ from Eqs. (A.3), (A.5), (A.10):

$$2(1 - \sin \alpha) \sin \beta + \cos \gamma = 1. \quad (\text{A.11})$$

Substitution of

$$\beta = \frac{\pi}{2} - 2\alpha \quad \text{and} \quad \beta = 3\alpha - \frac{\pi}{2} \quad (\text{A.12})$$

reduces Eq. (A.11) to

$$2 \cos 2\alpha + \sin \alpha = 1. \quad (\text{A.13})$$

The solution of Eq. (A.13) is

$$\sin \alpha = \sin \alpha_1 = \frac{\sqrt{17} + 1}{8}, \quad \alpha_1 \approx 39.820773^\circ. \quad (\text{A.14})$$

To find α_2 , consider rays traced in Fig. A1(b). Equations (A.2)–(A.7) are still valid. The following Eqs. (A.15)–(A.18) can be written instead of Eq. (A.8). From triangle BEI,

$$|\text{BD}| + |\text{DE}| = |\text{EI}| \tan \beta. \quad (\text{A.15})$$

From Fig. A1(b),

$$|\mathbf{EI}| = |\mathbf{EF}| + |\mathbf{FI}|, \quad (\text{A.16})$$

$$|\text{FI}| = R(1 \cos \delta), \quad (\text{A.17})$$

$$|\text{DE}| = R(\cos \alpha \sin \delta). \quad (\text{A.18})$$

Exclude $|DE|$, $|EI|$, and $|FI|$ from Eqs. (A.15)–(A.18):

$$|\text{BD}| = [|\text{EF}| + R(1 \cos \delta)] \tan \beta - R(\cos \alpha \sin \delta). \quad (\text{A.19})$$

Table A1Parameters of rays to calculate r_2 .

$\sin \alpha_2 = 0.82296097$	$\alpha_2 = 55.382307^\circ$	$\delta = 27.304904^\circ$
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Table A2

First two reflection factors for normal incidence on a close-packed row of circles.

$r_1 = \frac{\sqrt{17} + 1}{8} = 0.64038820$	$r_2 = 0.18257277$
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Exclude |DG| and |EF| from Eqs. (A.6), (A.7), (A.19):

$$|BD| = R(2 - \sin \alpha - \cos \delta) \tan \beta - R(\cos \alpha - \sin \delta). \quad (\text{A.20})$$

Exclude |AO| and |BD| from Eqs. (A.2), (A.4), (A.20):

$$|CD| = [(2 - \sin \alpha - \cos \delta) \sin \beta - (\cos \alpha - \sin \delta) \cos \beta] |CO|. \quad (\text{A.21})$$

Exclude |CO| and |CD| from Eqs. (A.3), (A.5), (A.21):

$$(2 - \sin \alpha - \cos \delta) \sin \beta - (\cos \alpha - \sin \delta) \cos \beta + \cos \gamma = 1. \quad (\text{A.22})$$

From triangle DEI,

$$\frac{|DE|}{|EI|} = \tan \varepsilon. \quad (\text{A.23})$$

Exclude |DE|, |DG|, |EF|, |EI| and |FI| from Eqs. (A.6), (A.7), (A.16)–(A.18), and (A.23):

$$(2 - \sin \alpha - \cos \delta) \sin \varepsilon = (\cos \alpha - \sin \delta) \cos \varepsilon. \quad (\text{A.24})$$

Substitution of

$$\varepsilon = 2\alpha - \frac{\pi}{2}, \quad \beta = 2\delta - \varepsilon, \quad \text{and} \quad \gamma = 3\alpha - 2\delta - \frac{\pi}{2} \quad (\text{A.25})$$

reduces Eqs. (A.22) and (A.24) to

$$(2 \cos 2\delta - \cos \delta) \cos 2\alpha + (2 \sin 2\delta - \sin \delta) \sin 2\alpha = 1, \quad (\text{A.26})$$

$$(2 - \sin \alpha - \cos \delta) \cos 2\alpha + (\cos \alpha - \sin \delta) \sin 2\alpha = 0. \quad (\text{A.27})$$

Numerical solution of the system of Eqs. (A.26), (A.27) gives the results shown in Table A1. Table A2 summarizes the reflection factors calculated by Eq. (A.1).

References

- [1] Wu H, Gui N, Yang X, Tu J, Jiang S. A matrix model of particle-scale radiative heat transfer in structured and randomly packed pebble bed. *Int J Thermal Sci* 2020;153:106334. doi:10.1016/j.ijthermalsci.2020.106334.
- [2] Ruiz G, Ripoll N, Fedorova N, Zbogar-Rasic A, Jovicic V, Delgado A, et al. Experimental and numerical analysis of the heat transfer in a packed bed exposed to the high thermal radiation flux. *Int J Heat Mass Transf* 2019;136:383–92.
- [3] Asakuma Y, Honda I, Yamamoto T. Numerical approach to predicting the effective thermal conductivity of a packed bed of binary particles. *Powder Technol* 2019;354:886–92.
- [4] Wang C-A, Ma L-X, Tan J-Y, Liu L-H. Study of radiative transfer in 1d densely packed bed layer containing absorbing/scattering spherical particles. *Int J Heat Mass Transf* 2016;102:669–78.
- [5] Johnson E, Tar I, Baker D. A Monte Carlo method to solve for radiative effective thermal conductivity for particle beds of various solid fractions and emissivities. *J Quant Spectrosc Radiat Transf* 2020;250:107014.
- [6] Kang HH, Kaya M, Hajimirza S. A data driven artificial neural network model for predicting radiative properties of metallic packed beds. *J Quant Spectrosc Radiat Transf* 2019;226:66–72.
- [7] Taine J, Enguehard F. Statistical modelling of radiative transfer within non-Bee-rian effective phases of macroporous media. *Int J Thermal Sci* 2019;139:61–78.
- [8] Tolochko NK, Laoui T, Khlopkov YV, Mozharov SE, Titov VI, Ignatiev MB. Absorptance of powder materials suitable for laser sintering. *Rapid Prototyping J* 2000;6(3):155–60.
- [9] Gusarov AV, Bentefour EH, Rombouts M, Froyen L, Glorieux C, Kruth J-P. Normal-directional and normal-hemispherical reflectances of micron- and submicron-sized powder beds at 633 and 790 nm. *J Appl Phys* 2006;99:113528.
- [10] Zhang D, Wang W, Guo Y, Hu S, Dong D, Poprawe R, et al. Numerical simulation in the absorption behavior of Ti6Al4V powder materials to laser energy during SLM. *J Mater Process Technol* 2019;268:25–36. doi:10.1016/j.jmatprotec.2019.01.002.
- [11] Boley CD, Mitchell SC, Rubenchik AM, Wu SSQ. Metal powder absorptivity: modeling and experiment. *Appl Opt* 2016;55(23):6496–500. doi:10.1364/AO.55.006496.
- [12] Meier C, Weissbach R, Weinberg J, Wall WA, John Hart A. Critical influences of particle size and adhesion on the powder layer uniformity in metal additive manufacturing. *J Mater Process Technol* 2019;266:484–501. doi:10.1016/j.jmatprotec.2018.10.037.
- [13] Meier C, Penny R, Zou Y, Gibbs J, John Hart A. Thermophysical phenomena in metal additive manufacturing by selective laser melting: fundamentals, modeling, simulation and experimentation. *Annu Rev Heat Transf* 2018;20:241–316.
- [14] Wang XC, Kruth J-P. A simulation model for direct selective laser sintering of metal powders. In: BHV Topping, editor. *Computational techniques for materials, composites and composite structures*. Edinburgh: Civil-Comp Press; 2000. p. 57–71.
- [15] Wang XC, Laoui T, Bonse J, Kruth J-P, Lauwers B, Froyen L. Direct selective laser sintering of hard metal powders: experimental study and simulation. *Int J Adv Manuf Technol* 2002;19(5):351–7.
- [16] Boley CD, Khairallah SA, Rubenchik AM. Calculation of laser absorption by metal powders in additive manufacturing. *Appl Opt* 2015;54(9):2477–82. doi:10.1364/AO.54.002477.
- [17] Gusarov AV, Kruth J-P. Modelling of radiation transfer in metallic powders at laser treatment. *Int J Heat Mass Transf* 2005;48(16):3423–34. doi:10.1016/j.ijheatmasstransfer.2005.01.044.
- [18] Gusarov AV, Smurov I. Radiation transfer in metallic powder beds used in laser processing. *J Quant Spectrosc Radiat Transf* 2010;111(17):2517–27. doi:10.1016/j.jqsrt.2010.07.009.
- [19] Rombouts M, Froyen L, Gusarov AV, Bentefour EH, Glorieux C. Light extinction in metallic powder beds: correlation with powder structure. *J Appl Phys* 2005;98:013533.
- [20] Kovalev OB, Gusarov AV. Modeling of granular packed beds, their statistical analyses and evaluation of effective thermal conductivity. *Int J Thermal Sci* 2017;114:327–41. doi:10.1016/j.ijthermalsci.2017.01.003.
- [21] Kovalev OB, Gusarov AV, Belyaev VV. Morphology of random packing of micro-particles and its effect on the absorption of laser radiation during selective melting of powders. *Int J Eng Sci* 2020;157:103378. doi:10.1016/j.jengsci.2020.103378.
- [22] Zarrouati M, Enguehard F, Taine J. Statistical characterization of near-wall radiative properties of a statistically non-homogeneous and anisotropic porous medium. *Int J Heat Mass Transf* 2013;67:776–83. doi:10.1016/j.ijheatmasstransfer.2013.08.021.
- [23] Khmyrov RS, Ableeva RR, Gusarov AV. Metallographic study of denudation in laser powder-bed fusion. *Proc CIRP* 2020;94:194–9. doi:10.1016/j.procir.2020.09.037.
- [24] Howell JR, Meng MP, Siegel R. *Thermal radiation heat transfer*. Taylor & Francis; 2015.
- [25] Gusarov AV. Statistical approach to radiative transfer in the heterogeneous media of thin-wall morphology-II: applications. *J Heat Transf* 2019;141:012701. doi:10.1115/1.4040958.